

Instructor: Date:

Learner: Regional Branch:

Program: Center:

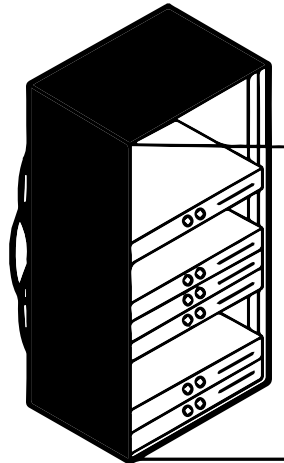
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Let's Start!

The World of Tools and Utilities



Learning Outcome:

Describe common activities at work to a visitor using short sentences in familiar spoken and written situations.
Understand information about data and its structure in simple spoken and written situations.

Ask your partner:

- How would you store the whole massive data of a company?
- Should information have a standard when you store it? Why?

Can Do Statements:

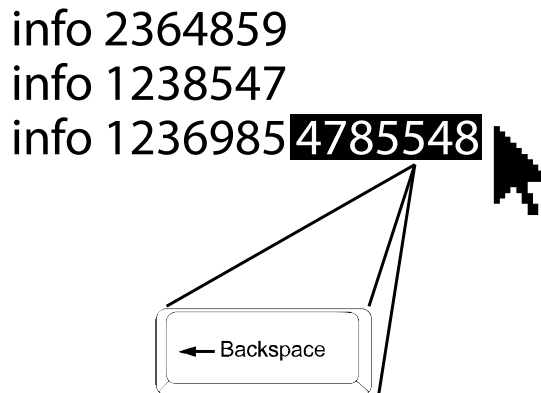
- I can talk about tools and utilities in data warehousing.
- I can give information about data extraction, data cleaning, and data transformation.
- I can explain key features related to the tools and utilities in data warehousing.





Let's Explore!

Tina is giving Tom a hand with a difficult topic. Can you help him to?



Instructions: Read the text below.

- Tom:** Hi, Tina. I need your help, please I need to know what is happening to our database. The sales manager and the CEO need some specific information from the database, but when I gather it from their computers, it is quite different.
- Tina:** Ok, I think I know what's happening. We need to implement a data warehouse to organize the whole data of the company.
- Tom:** I can convince the IT manager to do it if you tell me why it is important for the company.
- Tina:** Well, the thing is that when there are too many dependencies in a company, the information they have may differ from the others', and that complicates things for all departments.
- Tom:** Ok... how is data warehousing helping us in the company?
- Tina:** Well, that's easy. When we use data warehousing, it compiles all the data the company has from its different dependencies and databases. Then, when people need it, just look for it from their own offices by using the app of the company.
- Tom:** Hmm, but how does it manage the data?
- Tina:** Look, data warehouses have some tools and utilities that help the user through the process. For example, when you input a query to the program, it uses a data extraction tool to gather the data from the dependency you are requesting it from. Afterwards, the data cleaning tool corrects any possible errors that it finds in the databases. Once the app finds the information, it converts it by using the data transformation tool.
- There are some other important tools like the refreshing and data loading ones. Some key features of them are that they permit the data warehouse to update the data from the different databases and consolidate it, so that the data warehouse is always up to date.
- Tom:** Thanks for the advice Tina, let me see what I can do about it.



Let's Practice!

When do we need tools for data warehousing?

Instructions: Read the question. Select all the correct answers.

1. What happens when you use data warehousing for your company?
 - a. When you use data warehousing, all the data of your company goes to a huge building.
 - b. When you implement a data warehouse for you company, all the data that is there in the different dependencies of the company becomes accessible for everyone.
 - c. When a company has different databases in a company, a data warehouse is the best solution to that problem.
 - d. When your company has a data warehouse, they don't need to have any other database.
 - e. When a company implements data warehousing, it gathers, cleans and transforms all the data to put it there.
2. What is the data transformation tool of the data warehouse for?
 - a. When you implement the data transformation tool in the data warehouse, it gathers all the information of the company.
 - b. When the data warehouse implements the data transformation tool, it changes and modifies all the different data formats into one.
 - c. When you unify one single format for all the data you have in database to store it in the data warehouse, you are implementing the data transformation tool.
 - d. When the app implements the data transformation tool, it updates all the information that is there so that everyone can have access to updated information.



Let's Practice!

Take into account some key features!

Instructions: Select the correct answer to complete the sentence.

1. When you _____ the information of the database, you use the refreshing tool.
 - a. update
 - b. load
 - c. clean
2. When the system _____ the data into one single format, it is using the data transformation tool.
 - a. unifies
 - b. deletes
 - c. makes





Let's Practice!

3. When the system _____ the information from the different databases of the company, it uses the extraction tool.
 - a. brings
 - b. gathers
 - c. updates
4. When you make a _____ of all the data, the system is using the data loading tool.
 - a. consolidate
 - b. consolidating
 - c. consolidation
5. When the system _____ errors in the data and corrects them, it makes use of the data cleaning tool.
 - a. finds
 - b. deletes
 - c. fixes



Extension Activity

Now's your turn to show how much you know about data warehousing

Instructions: Complete the following steps. Fill in the Self-Assessment.

- a) Read about tools and utilities of data warehousing.
- b) Write a short paragraph describing all the tools and utilities of data management using the vocabulary of the lesson.
- c) Discuss your work with your partners.

Self-Assessment

Yes No

1. I read about tools and utilities of data warehousing.

☐ ☐

2. I write a short paragraph describing all the tools and utilities of data management using the vocabulary studied.

☐ ☐

3. I discuss my work with my partners.

☐ ☐



Target Words

- **CEO** (noun): Chief Executive Manager
e.g. The sales manager and the CEO need some specific information from the database, but when I gather it from their computers, it is quite different.
- **gather** (verb): to retrieve or collect something.
e.g. It uses a data extraction tool to gather the data from the dependency you are requesting it from.
- **input** (verb): to enter information into a text field or database
e.g. When you input a query to the program, it uses a data extraction tool to gather the data from the dependency you are requesting it from.
- **update** (verb): to keep informed with current information
e.g. They permit the data warehouse to update the data from the different databases and consolidate it, so that the data warehouse is always up to date.
- **manage** (verb): to control or direct
e.g. Hmm, but how does it manage the data?
- **convert** (verb): to change or modify
e.g. Once the app finds the information it converts it by using the data transformation tool.



Learning Points

Zero Conditional with When

Use **zero conditional** with **when** to express information about specific situations that commonly have a cause and effect.

In order to analyze the **zero conditional**, we must know what the **conditional mood** is.

We can see the conditional mood as a set of sentences showing the relation between facts (real or hypothetical) and their conditions.

Therefore, a conditional sentence has two clauses or ideas; the **main clause** (idea) or the effect and the **if clause** or conditional clause.

We can use the **zero conditional** to express ideas that are always true, that means factual ideas that always take place when something happens.

Both clauses or ideas in **zero conditional** have the verbs in present form; the conditional clause goes with "**if**" or "**when**," and the main clause goes in simple present.





Learning Points

Structure:

1. **If/when + present simple + , + present simple,**
2. **Present simple + if/when + present simple.**

Examples:

If you have problems with the databases, **create** a database warehouse.

Don't relay in one database for your company **if you want** to make things easy for the other dependencies.

Notice that when the **if/when** clause goes first, you need to add a comma “,” to separate both ideas; but when you start the idea with the main clause, you don't need to add a comma to separate the ideas.

Examples

- *The sales manager and the CEO need some specific information from the database, but when I gather it from their computers, it is quite different.*
- *When there are too many dependencies in a company, the information they have may differ from the others'.*
- *When we use data warehousing, it compiles all the data the company has from its different dependencies and databases.*

Subject and object pronouns

Use **object and subject pronouns** to replace nouns (names of people or things) so that you avoid redundancy in your text.

Pronouns are words that you can use to replace a noun in a sentence. Such **pronouns** can be either **object or subject pronouns**.

Compare:

James is creating a **database warehouse**. To do so, **James** needs all the dependencies give **James** what **James** needs to start building the **database warehouse**.

As you can see in the previous example, the writer repeats the nouns **James** and **Data warehouse** more than twice.





Learning Points

The correction of the previous paragraph should be like this:

James is creating a **database warehouse**. To do so, **he** needs all the dependencies give **him** what **he** needs to start building **it**.

Let's see the reason why:

SUBJECT PRONOUNS

The subject is the doer of the action, it can be a person, a thing or an animal. In a text you can use the subject pronoun to replace a previously mentioned subject so that you avoid redundancy.

In the previous example the writer repeats the noun **James** many times. In order to identify whether the noun is a subject or object pronoun you need to see where in the sentence it is located.

Subject + verb + object + complement.

The correction of the **subject pronouns** should look like follows.

James is creating a **database warehouse**. To do so, **he** needs all the dependencies give **James** what **he** needs to start building the **database warehouse**.

If you notice, the sentences **he needs all** and **he needs to** refer to **James** as the subject of the sentences; therefore, you can change it for "**he**."

Even though we fixed the redundancy regarding to the subject pronouns, there are still some repeated nouns.

OBJECT PRONOUNS

You use these pronouns to replace the object nouns in the sentence in order to avoid redundancy.

Remember that the object of the sentence is not an animal or inanimate thing, it is the what receives the action in a sentence.

Subject + verb + **object** + complement.

If we analyze the example, we realize that the repeated objects are **James** and **Warehouse**.

James is creating a **database warehouse**. To do so, he needs all the dependencies give **James** what he needs to start building the **database warehouse**.

The correct version should look like follows.





Learning Points

James is creating a **database warehouse**. To do so, he needs all the dependencies give **him** what he needs to start building **it**.

Look at the following chart to compare the subject and object pronouns.

Person	Subject Pronoun	Object Pronoun
1st singular	I	me
2nd singular	You	you
3rd singular	He	him
	She	her
	It	it
1st plural	We	us
2nd plural	You	you
3rd plural	They	them

Examples

- *I can convince the IT manager to do it if you tell me why it is important for the company.*
- *when people need it, just look for it from their own offices by using the app of the company.*
- *Once the app finds the information, it converts it by using the data transformation tool.*